

Preston Rank

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Education

University of Texas at Dallas

GPA: 3.57 | Bachelor of Science in Computer Science

Relevant Coursework: Software Engineering, Data Structures and Algorithms, Databases, Unix Programming

Richardson, TX

Expected Grad: Dec 2027

Uplift Summit International Preparatory High School

GPA: 3.88 | International Baccalaureate (IB) Diploma Programme

Arlington, TX

Experience

OrbitalMentorship

Software Engineer Intern

Richardson, TX

Aug 2025 – Jan 2026

- Delivered an **Express.js/React.js** analytics dashboard aggregating raw events into mentee progress metrics, directly improving feature prioritization decisions across **5+ educational modules**.
- Designed a **29+ entity MongoDB data model** to empower the analytics dashboard, providing on-demand visibility into session and engagement activity.
- Built a file processing system using **Express.js**, **AWS S3**, and **CloudFront CDN**, writing **unit tests** to validate upload and retrieval across assignments, resumes, and profiles.

ACM Research — (Capital of Texas Undergraduate Research Conference)

Truncated SVD-Based Prompt Compression for LLMs

Richardson, TX | Austin, TX

Aug 2025 – Mar 2026

- Integrated full prompt compression pipeline, compressing **Llama-2-7b** through **truncated SVD** then using **preserved subspaces** for **token scoring** and **pruning** to build the compressed prompt.
- Benchmarked inference against **LLMLingua2**, achieving **36.9% faster times** with only **9.2% cosine similarity reduction** for close-ended queries with SQuAD dataset, demonstrating a favorable efficiency gain.
- Presented research to a **judging panel** with varying technical backgrounds through a poster, including a **Q&A** portion that required defending technical decisions and explaining real world impact.

HackTX 2025 — (Toyota Track)

Car-ma!

Austin, TX

Oct 2025

- Shipped and demoed Car-ma, a dating-inspired Toyota car recommendation app, built the backend in **Express.js** and **DynamoDB**, deployed via **Vercel** in 24 hours.
- Collaborated with frontend and UI/UX teammates to deliver a complete full-stack experience for the Toyota track.

Projects

Internship Email Automation Agent Python CLI Tool

Dec 2025 – Mar 2026

- Built a **Dockerized** multi-agent CLI tool that automates personalized internship outreach emails through an FSM-orchestrated pipeline integrated with the Gmail API.
- Designed a normalized **PostgreSQL schema** and **DDL script**, tracking companies, recruiters, and emails, generating insights like response rates, company coverage, and follow-up candidates.
- Coordinated **Pydantic AI agents** for email personalization and resume selection using role and company specific context, with rate-limiting and caching to handle API constraints reliably.

iMessage RAG Query System with Conversation Disentanglement ML Model

Oct 2025 – Jan 2026

- Built a **Python RAG app** allowing natural language search over personal iMessage history, with contact-based extraction from the macOS **SQLite** database and a **Chainlit** chat interface.
- Developed a three-stage pipeline to chunk raw messages into coherent conversations using time, semantic similarity, and an ML reconstruction model, storing embeddings in **FAISS** for retrieval.
- Trained a self-attention model with dual-head ranking and classification for conversation disentanglement, achieving **74.2% F1 score** on IRC Disentanglement dataset.

Thrift Store Discovery App

Jun 2025 – Aug 2025

- Developed an iOS app that surfaces and categorizes nearby thrift stores beyond Google Maps, using **SwiftUI** and **MapKit** with direct filters for style, type, and budget.
- Built using a Node.js/Express backend integrating GPT-4 and Google Places to automatically categorize store listings from metadata, with **MongoDB** and **AWS S3** handling persistent store data and photos.

Technical Skills

Languages: Python, JavaScript, TypeScript, Java, C++, C, SQL, Swift

Frameworks: Express.js, React.js, PyTorch, Hugging Face Transformers, FAISS, SwiftUI

Tools and Technologies: Git, Docker, AWS, PostgreSQL, MongoDB, DynamoDB, SQLite, Vercel, CI/CD